COMP3153/9153

Algorithmic Verification

Lecture 11: Abstract Interpretation and Static Analysis

Static Analysis

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Most tools aren't sound nor complete, because they're a mixture.

Abstract Interpretation

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We have seen this before. Predicate abstraction is an example of abstract interpretation.

The WHILE Language

To make things easier, we will define a simple imperative programming language as follows:

$$
\begin{array}{rcl}
\mathcal{A} & ::= & \langle \textit{arithmetic expressions} \rangle \\
\mathcal{B} & ::= & \langle \textit{boolean expressions} \rangle \\
\mathcal{S} & ::= & [x := \mathcal{A}]^{\ell} \mid [\textbf{skip}]^{\ell} \mid \mathcal{S}_1 ; \mathcal{S}_2 \\
 & \mid & \textbf{if } [\mathcal{B}]^{\ell} \textbf{ then } \mathcal{S} \textbf{ else } \mathcal{S} \\
 & \mid & \textbf{while } [\mathcal{B}]^{\ell} \textbf{ do } \mathcal{S}
\end{array}
$$

Note that we *label* all statements and conditions (usually with a number). These labelled terms correspond to nodes on the control flow graph.

Examples

$$
\begin{array}{l}
\textbf{if } [x > 0]^1 \textbf{ then} \\
\quad [x := 2x + 1]^2 \\
\textbf{else} \\
\quad [x := 1 - 4x]^3 \\
[x := 8 \div x]^4
\end{array}
$$

Examples

if $[x > 0]^1$ **then**
    $[x := 2x + 1]^2$
else
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What abstract domain should we use to detect divide by zero?

Intervals

Let's define intervals n, m as either:

- $\emptyset$, the empty interval, or
- $[n, m]$ where $n, m \in \mathbb{Z} \cup \{+\infty, -\infty\}$.

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Observation

Define $\inf S = \bigcap S$ and $\sup S = \bigcup S$, then intervals form a lattice: $\bot = \emptyset$ and $\top = [-\infty, +\infty]$. The ordering $\preceq$ here is interval inclusion.

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We can "lift" arithmetic operators to the interval level, where they apply to both bounds. Similarly define e.g. $\hat{3} = [3, 3]$.

Equation Systems

We define a series of *entry* interval equations x_i, and a series of *exit* equations x_i' describing the possible values of x before and after the statement i.

$$x_1 \quad =$$

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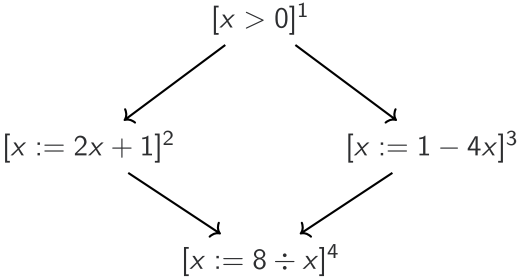

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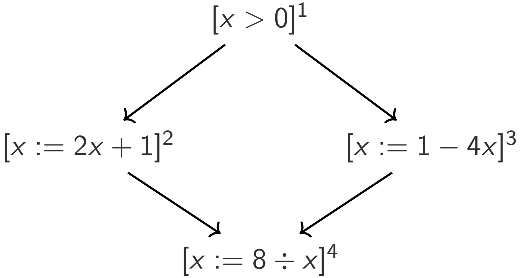

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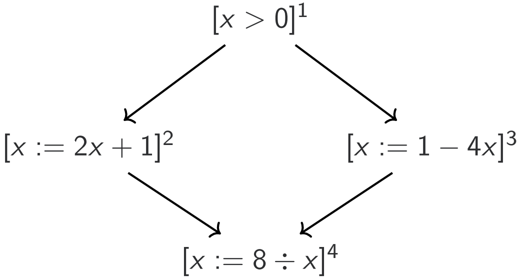

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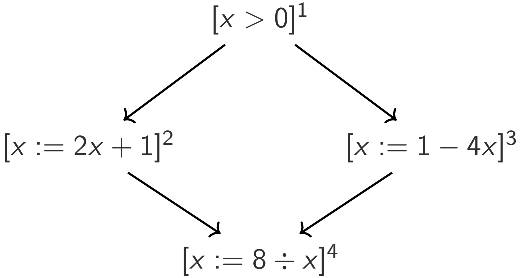

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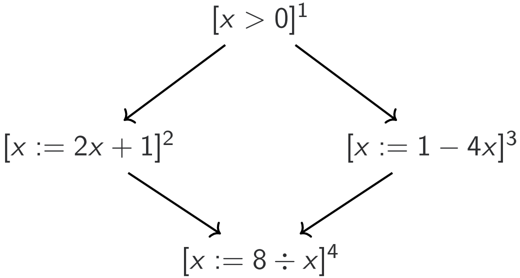

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Observe that all operations used are monotone. How can we compute what the intervals are? Iterate to the least fixed point !

Least Fixed Points for Equation Systems

We start initialising all equations to $\perp$, and then iterate until the results stop changing. We can choose the equations in any (fair) order, and we will always reach a fixed point eventually. Some ways are faster than others.

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Seeing as $0 \notin x_4$, we know divide by zero is impossible.

Slow Convergence

Because the previous example had no loops, all equations converged after one step. Compare to this example:

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[n := 1]^1 \\
\textbf{while } [n < 1000]^2 \textbf{ do} \\
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[\textbf{skip}]^4
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$$

Slightly simplified equations for presentation:

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n'_1 & = & \emptyset & = & [1, 1] \\
n_2 & = & \emptyset & = & n'_1 \cup n'_3 \\
n_3 & = & \emptyset & = & n_2 \cap [-\infty, 999] \\
n'_3 & = & \emptyset & = & n_3 + 1 \\
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This is going to take a long time to converge! (1000 steps)

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Our interval abstraction is too detailed, making our loop iterations take ages.

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Let n be the value we are updating and m be the result of the next iteration. Then, we update with $n \triangledown m$ instead of m:

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\emptyset \triangledown m & = & m \\
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[\ell_0, u_0] \triangledown [\ell_1, u_1] & = & [\textbf{if } \ell_1 < \ell_0 \textbf{ then } -\infty \textbf{ else } \ell_1 \\
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In other words, if we ever try to loosen a bound, we just extrapolate all the way to infinity.

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This is an overapproximation, but it converges much faster than the normal iterative sequence does.

Loops with Widening

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\begin{array}{lclcl}
n_1' & = & \emptyset & = & [1,1] \\
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Beyond Interval Anaylsis

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Interval analysis is very effective but not very accurate, because it doesn't express the relationships between variables.
Predicate abstraction and *polyhedral models* do better in many cases, but are more complicated.
All are based on the same principle of least-fixed point of a system of equations.

Data-flow Analysis

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Data-flow analyses may be *forwards* or *backwards*, and *may* or *must*.
AEA is a forwards must analysis. LVA is a backwards may analysis.

Step 1: Gen and Kill

Each location in the CFG has an associated gen set, of generated information, and kill set, of information that is no longer accurate.

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In AEA, $\text{gen}_{AE}(\ell)$ is the expressions evaluated (and not updated) in ℓ and $\text{kill}_{AE}(\ell)$ is those expressions updated by ℓ.

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Note: $a := a + 1$ would kill $a + 1$, not generate it. Why?

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Example (LVA)

In LVA, $\text{gen}_{LV}(\ell)$ is the variables read (and not written to) in ℓ and $\text{kill}_{LV}(\ell)$ would be the variables written to in ℓ.
For example, $x := a + b$ would generate $\{a, b\}$ and kill $\{x\}$.

Available Expressions Example

$$
\begin{tikzcd}
{[x := a + b]^1} \arrow[d] \\
{[y := a * b]^2} \arrow[d] \\
{[y > a + b]^3} \arrow[d] \arrow[r] & {} \\
{[a := a + 1]^4} \arrow[d] \\
{[x := a + b]^5} \arrow[uu, bend left]
\end{tikzcd}
$$

ℓ	gen_{AE}	kill_{AE}
1		

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1	$\{a + b\}$	

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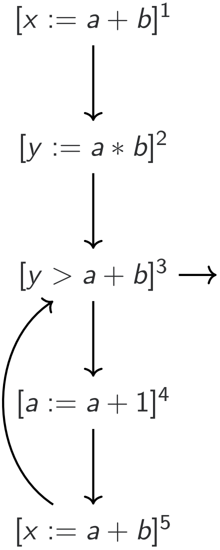
\end{tikzcd}
$$

ℓ	gen_{AE}	kill_{AE}
1	$\{a + b\}$	$\emptyset$
2		

Available Expressions Example



ℓ	gen_{AE}	kill_{AE}
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2	$\{a * b\}$	$\emptyset$
3		

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$$

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1	$\{a + b\}$	$\emptyset$
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5		

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4	$\emptyset$	$\{a + b, a * b, a + 1\}$
5	$\{a + b\}$	$\emptyset$

What to do now?

Specify an equation system that relates these, then find the fixed point!

Forwards Must Analysis

The set of available expressions as we enter a location ℓ is the intersection of all available expressions from the predecessors to ℓ:

$$AE_{\text{entry}}(\ell) = \begin{cases} \emptyset & \text{if } \ell \in I_{\text{CFG}} \\ \bigcap\{AE_{\text{exit}}(\ell') \mid (\ell', \ell) \in \delta_{\text{CFG}}\} & \text{otherwise} \end{cases}$$

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The set of available expressions as we enter a location ℓ is the intersection of all available expressions from the predecessors to ℓ:

$$AE_{\text{entry}}(\ell) = \begin{cases} \emptyset & \text{if } \ell \in I_{\text{CFG}} \\ \bigcap\{AE_{\text{exit}}(\ell') \mid (\ell', \ell) \in \delta_{\text{CFG}}\} & \text{otherwise} \end{cases}$$

We choose intersection because we want expressions that are **definitely** available no matter what route we took to get to ℓ (a **must** analysis).

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We choose intersection because we want expressions that are **definitely** available no matter what route we took to get to ℓ (a **must** analysis).

The set of available expressions as we exit a location ℓ is the set that we entered with, minus anything we kill, plus anything we generate:

$$AE_{\text{exit}}(\ell) = (AE_{\text{entry}}(\ell) \setminus \text{kill}_{\text{AE}}(\ell)) \cup \text{gen}_{\text{AE}}(\ell)$$

Example for AEA

$$
\begin{tikzcd}
{[x := a + b]^1} \arrow[d] \\
{[y := a * b]^2} \arrow[d] \\
{[y > a + b]^3} \arrow[d] \\
{[a := a + 1]^4} \arrow[d] \\
{[x := a + b]^5} \arrow[uu, bend left]
\end{tikzcd}
$$

ℓ	gen_{AE}	kill_{AE}
1	$\{a + b\}$	$\emptyset$
2	$\{a * b\}$	$\emptyset$
3	$\{a + b\}$	$\emptyset$
4	$\emptyset$	$\{a + b, a * b, a + 1\}$
5	$\{a + b\}$	

ℓ	$AE_{\text{entry}}(\ell)$	$AE_{\text{exit}}(\ell)$
1	$\emptyset$	

Example for AEA

$$
\begin{tikzcd}
{[x := a+b]^1} \arrow[d] \\
{[y := a*b]^2} \arrow[d] \\
{[y > a+b]^3} \arrow[d] \arrow[r] & {} \\
{[a := a+1]^4} \arrow[d] \\
{[x := a+b]^5} \arrow[uu, bend left=60]
\end{tikzcd}
$$

ℓ	gen_{AE}	$kill_{AE}$
1	$\{a+b\}$	$\emptyset$
2	$\{a*b\}$	$\emptyset$
3	$\{a+b\}$	$\emptyset$
4	$\emptyset$	$\{a+b, a*b, a+1\}$
5	$\{a+b\}$	

ℓ	$AE_{entry}(\ell)$	$AE_{exit}(\ell)$
1	$\emptyset$	$AE_{entry}(1) \cup \{a+b\}$

Example for AEA

$$
\begin{tikzcd}
{[x := a+b]^1} \arrow[d] \\
{[y := a*b]^2} \arrow[d] \\
{[y > a+b]^3} \arrow[d] \arrow[r] & {} \\
{[a := a+1]^4} \arrow[d] \\
{[x := a+b]^5} \arrow[uu, bend left]
\end{tikzcd}
$$

ℓ	gen_{AE}	$kill_{AE}$
1	$\{a+b\}$	$\emptyset$
2	$\{a*b\}$	$\emptyset$
3	$\{a+b\}$	$\emptyset$
4	$\emptyset$	$\{a+b, a*b, a+1\}$
5	$\{a+b\}$	

ℓ	$AE_{entry}(\ell)$	$AE_{exit}(\ell)$
1	$\emptyset$	$AE_{entry}(1) \cup \{a+b\}$
2	$AE_{exit}(1)$	

Example for AEA

$$
\begin{tikzcd}
{[x := a + b]^1} \arrow[d] \\
{[y := a * b]^2} \arrow[d] \\
{[y > a + b]^3} \arrow[d] \arrow[r] & {} \\
{[a := a + 1]^4} \arrow[d] \\
{[x := a + b]^5} \arrow[uu, bend left]
\end{tikzcd}
$$

ℓ	gen_{AE}	kill_{AE}
1	$\{a + b\}$	$\emptyset$
2	$\{a * b\}$	$\emptyset$
3	$\{a + b\}$	$\emptyset$
4	$\emptyset$	$\{a + b, a * b, a + 1\}$
5	$\{a + b\}$	

ℓ	$AE_{\text{entry}}(\ell)$	$AE_{\text{exit}}(\ell)$
1	$\emptyset$	$AE_{\text{entry}}(1) \cup \{a + b\}$
2	$AE_{\text{exit}}(1)$	$AE_{\text{entry}}(2) \cup \{a * b\}$

Example for AEA

$$
\begin{tikzcd}
{[x := a + b]^1} \arrow[d] \\
{[y := a * b]^2} \arrow[d] \\
{[y > a + b]^3} \arrow[d] \arrow[r] & {} \\
{[a := a + 1]^4} \arrow[d] \\
{[x := a + b]^5} \arrow[uu, bend left]
\end{tikzcd}
$$

ℓ	gen_{AE}	kill_{AE}
1	$\{a + b\}$	$\emptyset$
2	$\{a * b\}$	$\emptyset$
3	$\{a + b\}$	$\emptyset$
4	$\emptyset$	$\{a + b, a * b, a + 1\}$
5	$\{a + b\}$	

ℓ	$AE_{\text{entry}}(\ell)$	$AE_{\text{exit}}(\ell)$
1	$\emptyset$	$AE_{\text{entry}}(1) \cup \{a + b\}$
2	$AE_{\text{exit}}(1)$	$AE_{\text{entry}}(2) \cup \{a * b\}$
3	$AE_{\text{exit}}(2) \cap AE_{\text{exit}}(5)$	

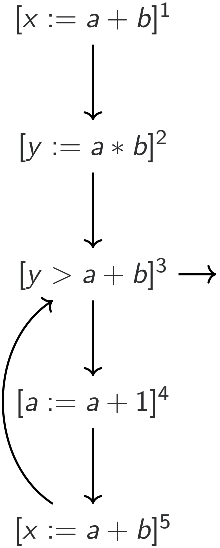
Example for AEA

$$
\begin{tikzcd}
{[x := a + b]^1} \arrow[d] \\
{[y := a * b]^2} \arrow[d] \\
{[y > a + b]^3} \arrow[d] \arrow[r] & {} \\
{[a := a + 1]^4} \arrow[d] \\
{[x := a + b]^5} \arrow[uu, bend left]
\end{tikzcd}
$$

ℓ	gen_{AE}	$kill_{AE}$
1	$\{a+b\}$	$\emptyset$
2	$\{a*b\}$	$\emptyset$
3	$\{a+b\}$	$\emptyset$
4	$\emptyset$	$\{a+b, a*b, a+1\}$
5	$\{a+b\}$	

ℓ	$AE_{entry}(\ell)$	$AE_{exit}(\ell)$
1	$\emptyset$	$AE_{entry}(1) \cup \{a+b\}$
2	$AE_{exit}(1)$	$AE_{entry}(2) \cup \{a*b\}$
3	$AE_{exit}(2) \cap AE_{exit}(5)$	$AE_{entry}(3) \cup \{a+b\}$

Example for AEA



ℓ	gen_{AE}	kill_{AE}
1	$\{a + b\}$	$\emptyset$
2	$\{a * b\}$	$\emptyset$
3	$\{a + b\}$	$\emptyset$
4	$\emptyset$	$\{a + b, a * b, a + 1\}$
5	$\{a + b\}$	

ℓ	$AE_{\mathrm{entry}}(\ell)$	$AE_{\mathrm{exit}}(\ell)$
1	$\emptyset$	$AE_{\mathrm{entry}}(1) \cup \{a + b\}$
2	$AE_{\mathrm{exit}}(1)$	$AE_{\mathrm{entry}}(2) \cup \{a * b\}$
3	$AE_{\mathrm{exit}}(2) \cap AE_{\mathrm{exit}}(5)$	$AE_{\mathrm{entry}}(3) \cup \{a + b\}$
4	$AE_{\mathrm{exit}}(3)$	

Example for AEA

$$
\begin{tikzcd}
{[x := a + b]^1} \arrow[d] \\
{[y := a * b]^2} \arrow[d] \\
{[y > a + b]^3} \arrow[d] \arrow[r] & {} \\
{[a := a + 1]^4} \arrow[d] \\
{[x := a + b]^5} \arrow[uu, bend left]
\end{tikzcd}
$$

ℓ	gen_{AE}	$kill_{AE}$
1	$\{a+b\}$	$\emptyset$
2	$\{a*b\}$	$\emptyset$
3	$\{a+b\}$	$\emptyset$
4	$\emptyset$	$\{a+b, a*b, a+1\}$
5	$\{a+b\}$	

ℓ	$AE_{entry}(\ell)$	$AE_{exit}(\ell)$
1	$\emptyset$	$AE_{entry}(1) \cup \{a+b\}$
2	$AE_{exit}(1)$	$AE_{entry}(2) \cup \{a*b\}$
3	$AE_{exit}(2) \cap AE_{exit}(5)$	$AE_{entry}(3) \cup \{a+b\}$
4	$AE_{exit}(3)$	$AE_{entry}(4) \setminus kill_{AE}(4)$

Example for AEA

$$
\begin{tikzcd}
{[x := a + b]^1} \arrow[d] \\
{[y := a * b]^2} \arrow[d] \\
{[y > a + b]^3} \arrow[d] \\
{[a := a + 1]^4} \arrow[d] \\
{[x := a + b]^5} \arrow[uu, bend left]
\end{tikzcd}
\longrightarrow
$$

ℓ	gen_{AE}	kill_{AE}
1	$\{a+b\}$	$\emptyset$
2	$\{a*b\}$	$\emptyset$
3	$\{a+b\}$	$\emptyset$
4	$\emptyset$	$\{a+b, a*b, a+1\}$
5	$\{a+b\}$	

ℓ	$AE_{\text{entry}}(\ell)$	$AE_{\text{exit}}(\ell)$
1	$\emptyset$	$AE_{\text{entry}}(1) \cup \{a+b\}$
2	$AE_{\text{exit}}(1)$	$AE_{\text{entry}}(2) \cup \{a*b\}$
3	$AE_{\text{exit}}(2) \cap AE_{\text{exit}}(5)$	$AE_{\text{entry}}(3) \cup \{a+b\}$
4	$AE_{\text{exit}}(3)$	$AE_{\text{entry}}(4) \setminus \text{kill}_{AE}(4)$
5	$AE_{\text{exit}}(4)$	

Example for AEA

$$
\begin{tikzcd}
{[x := a + b]^1} \arrow[d] \\
{[y := a * b]^2} \arrow[d] \\
{[y > a + b]^3} \arrow[d] \arrow[r] & {} \\
{[a := a + 1]^4} \arrow[d] \\
{[x := a + b]^5} \arrow[uu, bend left]
\end{tikzcd}
$$

ℓ	gen_{AE}	$kill_{AE}$
1	$\{a + b\}$	$\emptyset$
2	$\{a * b\}$	$\emptyset$
3	$\{a + b\}$	$\emptyset$
4	$\emptyset$	$\{a + b, a * b, a + 1\}$
5	$\{a + b\}$	

ℓ	$AE_{\text{entry}}(\ell)$	$AE_{\text{exit}}(\ell)$
1	$\emptyset$	$AE_{\text{entry}}(1) \cup \{a + b\}$
2	$AE_{\text{exit}}(1)$	$AE_{\text{entry}}(2) \cup \{a * b\}$
3	$AE_{\text{exit}}(2) \cap AE_{\text{exit}}(5)$	$AE_{\text{entry}}(3) \cup \{a + b\}$
4	$AE_{\text{exit}}(3)$	$AE_{\text{entry}}(4) \setminus kill_{AE}(4)$
5	$AE_{\text{exit}}(4)$	$AE_{\text{entry}}(5) \cup \{a + b\}$

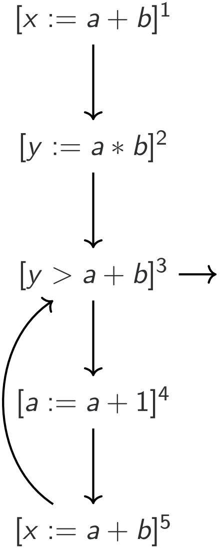
Results

$$
\begin{tikzcd}
{[x := a + b]^1} \arrow[d] \\
{[y := a * b]^2} \arrow[d] \\
{[y > a + b]^3} \arrow[d] \arrow[r] & {} \\
{[a := a + 1]^4} \arrow[d] \\
{[x := a + b]^5} \arrow[uu, bend left]
\end{tikzcd}
$$

ℓ	gen_{AE}	kill_{AE}
1	$\{a + b\}$	$\emptyset$
2	$\{a * b\}$	$\emptyset$
3	$\{a + b\}$	$\emptyset$
4	$\emptyset$	$\{a + b, a * b, a + 1\}$
5	$\{a + b\}$	

ℓ	$AE_{\text{entry}}(\ell)$	$AE_{\text{exit}}(\ell)$
1	$\emptyset$	$\{a + b\}$
2	$\{a + b\}$	$\{a + b, a * b\}$
3	$\{a + b\}$	$\{a + b\}$
4	$\{a + b\}$	$\emptyset$
5	$\emptyset$	$\{a + b\}$

Results



ℓ	gen_{AE}	$kill_{AE}$
1	$\{a + b\}$	$\emptyset$
2	$\{a * b\}$	$\emptyset$
3	$\{a + b\}$	$\emptyset$
4	$\emptyset$	$\{a + b, a * b, a + 1\}$
5	$\{a + b\}$	

ℓ	$AE_{entry}(\ell)$	$AE_{exit}(\ell)$
1	$\emptyset$	$\{a + b\}$
2	$\{a + b\}$	$\{a + b, a * b\}$
3	$\{a + b\}$	$\{a + b\}$
4	$\{a + b\}$	$\emptyset$
5	$\emptyset$	$\{a + b\}$

Note $\ell = 3$ can be optimised, as $a + b$ is already computed!

Backwards May Analysis

The set of live variables as we exit a location ℓ is the union of live variables from the successors to ℓ:

$$LV_{\text{exit}}(\ell) = \begin{cases} \emptyset & \text{if } \ell \in F_{\text{CFG}} \\ \bigcup\{LV_{\text{entry}}(\ell') \mid (\ell, \ell') \in \delta_{\text{CFG}}\} & \text{otherwise} \end{cases}$$

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We choose union because we want any variables that **might be** used in a successor to be marked live in ℓ (a **may** analysis).

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We choose union because we want any variables that **might be** used in a successor to be marked live in ℓ (a **may** analysis).

The set of live variables as we enter a location ℓ is the set of live variables at the exit, minus anything we kill (write to), plus anything we generate (read from):

$$LV_{\text{entry}}(\ell) = (LV_{\text{exit}}(\ell) \setminus \text{kill}_{\text{LV}}(\ell)) \cup \text{gen}_{\text{LV}}(\ell)$$

Backwards May Analysis

The set of live variables as we exit a location ℓ is the union of live variables from the successors to ℓ:

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The set of live variables as we enter a location ℓ is the set of live variables at the exit, minus anything we kill (write to), plus anything we generate (read from):

$$LV_{\text{entry}}(\ell) = (LV_{\text{exit}}(\ell) \setminus \text{kill}_{\text{LV}}(\ell)) \cup \text{gen}_{\text{LV}}(\ell)$$

The entry and exit equations are flipped because this is a backwards analysis.

Example for LVA

$$
\begin{tikzcd}
{[a := 0]^1} \arrow[d] \\
{[b := a * 1]^2} \arrow[d] \\
{[c := c + b]^3} \arrow[d] \\
{[a := b * 2]^4} \arrow[d] \\
{[a < 9]^5} \arrow[r] \arrow[uuu, bend left] & {[\textbf{return } c]^6}
\end{tikzcd}
$$

ℓ	gen_{LV}	kill_{LV}
1		

Example for LVA

$$
\begin{tikzcd}
{[a := 0]^1} \arrow[d] \\
{[b := a * 1]^2} \arrow[d] \\
{[c := c + b]^3} \arrow[d] \\
{[a := b * 2]^4} \arrow[d] \\
{[a < 9]^5} \arrow[r] \arrow[uuu, bend left] & {[\textbf{return } c]^6}
\end{tikzcd}
$$

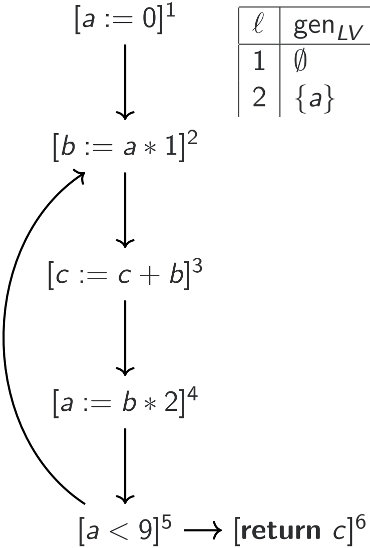
ℓ	gen_{LV}	kill_{LV}
1	$\emptyset$	

Example for LVA

$$
\begin{tikzcd}
{[a := 0]^1} \arrow[d] & \\
{[b := a * 1]^2} \arrow[d] & \\
{[c := c + b]^3} \arrow[d] & \\
{[a := b * 2]^4} \arrow[d] & \\
{[a < 9]^5} \arrow[r] \arrow[uuu, bend left] & {[\textbf{return } c]^6}
\end{tikzcd}
$$

ℓ	gen_{LV}	kill_{LV}
1	$\emptyset$	$\{a\}$
2		

Example for LVA



ℓ	gen_{LV}	kill_{LV}
1	$\emptyset$	$\{a\}$
2	$\{a\}$	

Example for LVA

$$
\begin{tikzcd}
{[a := 0]^1} \arrow[d] & \\
{[b := a * 1]^2} \arrow[d] & \\
{[c := c + b]^3} \arrow[d] & \\
{[a := b * 2]^4} \arrow[d] & \\
{[a < 9]^5} \arrow[uuu, bend left=60] \arrow[r] & {[\textbf{return } c]^6}
\end{tikzcd}
$$

ℓ	gen_{LV}	kill_{LV}
1	$\emptyset$	$\{a\}$
2	$\{a\}$	$\{b\}$
3		

Example for LVA

$$
\begin{tikzcd}
{[a := 0]^1} \arrow[d] & \\
{[b := a * 1]^2} \arrow[d] & \\
{[c := c + b]^3} \arrow[d] & \\
{[a := b * 2]^4} \arrow[d] & \\
{[a < 9]^5} \arrow[r] \arrow[uuu, bend left=60] & {[\textbf{return } c]^6}
\end{tikzcd}
$$

ℓ	gen_{LV}	kill_{LV}
1	$\emptyset$	$\{a\}$
2	$\{a\}$	$\{b\}$
3	$\{b\}$	$\{c\}$
4		

Example for LVA

$$
\begin{tikzcd}
{[a := 0]^1} \arrow[d] & \\
{[b := a * 1]^2} \arrow[d] & \\
{[c := c + b]^3} \arrow[d] & \\
{[a := b * 2]^4} \arrow[d] & \\
{[a < 9]^5} \arrow[r] \arrow[uuu, bend left=60] & {[\textbf{return } c]^6}
\end{tikzcd}
$$

ℓ	gen_{LV}	kill_{LV}
1	$\emptyset$	$\{a\}$
2	$\{a\}$	$\{b\}$
3	$\{b\}$	$\{c\}$
4	$\{b\}$	$\{a\}$
5		

Example for LVA

$$
\begin{tikzcd}
{[a := 0]^1} \arrow[d] & \\
{[b := a * 1]^2} \arrow[d] & \\
{[c := c + b]^3} \arrow[d] & \\
{[a := b * 2]^4} \arrow[d] & \\
{[a < 9]^5} \arrow[r] \arrow[uuu, bend left] & {[\textbf{return } c]^6}
\end{tikzcd}
$$

ℓ	gen_{LV}	kill_{LV}
1	$\emptyset$	$\{a\}$
2	$\{a\}$	$\{b\}$
3	$\{b\}$	$\{c\}$
4	$\{b\}$	$\{a\}$
5	$\{a\}$	$\emptyset$

ℓ	$LV_{\text{entry}}(\ell)$	$LV_{\text{exit}}(\ell)$
1		

Example for LVA

$$
\begin{tikzcd}
{[a := 0]^1} \arrow[d] & \\
{[b := a * 1]^2} \arrow[d] & \\
{[c := c + b]^3} \arrow[d] & \\
{[a := b * 2]^4} \arrow[d] & \\
{[a < 9]^5} \arrow[r] \arrow[uuu, bend left] & {[\textbf{return } c]^6}
\end{tikzcd}
$$

ℓ	gen_{LV}	kill_{LV}
1	$\emptyset$	$\{a\}$
2	$\{a\}$	$\{b\}$
3	$\{b\}$	$\{c\}$
4	$\{b\}$	$\{a\}$
5	$\{a\}$	$\emptyset$

ℓ	$LV_{\text{entry}}(\ell)$	$LV_{\text{exit}}(\ell)$
1	$LV_{\text{exit}}(1) \setminus \{a\}$	

Example for LVA

$$\begin{tikzcd}
{[a := 0]^1} \arrow[d] \\
{[b := a * 1]^2} \arrow[d] \\
{[c := c + b]^3} \arrow[d] \\
{[a := b * 2]^4} \arrow[d] \\
{[a < 9]^5} \arrow[r] \arrow[uuu, bend left] & {[\textbf{return } c]^6}
\end{tikzcd}$$

ℓ	gen_{LV}	kill_{LV}
1	$\emptyset$	$\{a\}$
2	$\{a\}$	$\{b\}$
3	$\{b\}$	$\{c\}$
4	$\{b\}$	$\{a\}$
5	$\{a\}$	$\emptyset$

ℓ	$LV_{\text{entry}}(\ell)$	$LV_{\text{exit}}(\ell)$
1	$LV_{\text{exit}}(1) \setminus \{a\}$	$LV_{\text{entry}}(2)$
2		

Example for LVA

$$
\begin{tikzcd}
{[a := 0]^1} \arrow[d] \\
{[b := a * 1]^2} \arrow[d] \\
{[c := c + b]^3} \arrow[d] \\
{[a := b * 2]^4} \arrow[d] \\
{[a < 9]^5} \arrow[uuu, bend left] \arrow[r] & {[\textbf{return } c]^6}
\end{tikzcd}
$$

ℓ	gen_{LV}	kill_{LV}
1	$\emptyset$	$\{a\}$
2	$\{a\}$	$\{b\}$
3	$\{b\}$	$\{c\}$
4	$\{b\}$	$\{a\}$
5	$\{a\}$	$\emptyset$

ℓ	$LV_{\text{entry}}(\ell)$	$LV_{\text{exit}}(\ell)$
1	$LV_{\text{exit}}(1) \setminus \{a\}$	$LV_{\text{entry}}(2)$
2	$LV_{\text{exit}}(2) \setminus \{b\} \cup \{a\}$	

Example for LVA

$$
\begin{tikzcd}
{[a := 0]^1} \arrow[d] \\
{[b := a * 1]^2} \arrow[d] \\
{[c := c + b]^3} \arrow[d] \\
{[a := b * 2]^4} \arrow[d] \\
{[a < 9]^5} \arrow[r] \arrow[uuu, bend left] & {[\textbf{return } c]^6}
\end{tikzcd}
$$

ℓ	gen_{LV}	kill_{LV}
1	$\emptyset$	$\{a\}$
2	$\{a\}$	$\{b\}$
3	$\{b\}$	$\{c\}$
4	$\{b\}$	$\{a\}$
5	$\{a\}$	$\emptyset$

ℓ	$LV_{\text{entry}}(\ell)$	$LV_{\text{exit}}(\ell)$
1	$LV_{\text{exit}}(1) \setminus \{a\}$	$LV_{\text{entry}}(2)$
2	$LV_{\text{exit}}(2) \setminus \{b\} \cup \{a\}$	$LV_{\text{entry}}(3)$
3		

Example for LVA

$$
\begin{tikzcd}
{[a := 0]^1} \arrow[d] \\
{[b := a * 1]^2} \arrow[d] \\
{[c := c + b]^3} \arrow[d] \\
{[a := b * 2]^4} \arrow[d] \\
{[a < 9]^5} \arrow[r] \arrow[uuu, bend left] & {[\textbf{return } c]^6}
\end{tikzcd}
$$

ℓ	gen_{LV}	kill_{LV}
1	$\emptyset$	$\{a\}$
2	$\{a\}$	$\{b\}$
3	$\{b\}$	$\{c\}$
4	$\{b\}$	$\{a\}$
5	$\{a\}$	$\emptyset$

ℓ	$LV_{\text{entry}}(\ell)$	$LV_{\text{exit}}(\ell)$
1	$LV_{\text{exit}}(1) \setminus \{a\}$	$LV_{\text{entry}}(2)$
2	$LV_{\text{exit}}(2) \setminus \{b\} \cup \{a\}$	$LV_{\text{entry}}(3)$
3	$LV_{\text{exit}}(3) \setminus \{c\} \cup \{b\}$	

Example for LVA

$$
\begin{tikzcd}
{[a := 0]^1} \arrow[d] & \\
{[b := a * 1]^2} \arrow[d] & \\
{[c := c + b]^3} \arrow[d] & \\
{[a := b * 2]^4} \arrow[d] & \\
{[a < 9]^5} \arrow[r] \arrow[uuu, bend left=60] & {[\textbf{return } c]^6}
\end{tikzcd}
$$

ℓ	gen_{LV}	$kill_{LV}$
1	$\emptyset$	$\{a\}$
2	$\{a\}$	$\{b\}$
3	$\{b\}$	$\{c\}$
4	$\{b\}$	$\{a\}$
5	$\{a\}$	$\emptyset$

ℓ	$LV_{entry}(\ell)$	$LV_{exit}(\ell)$
1	$LV_{exit}(1) \setminus \{a\}$	$LV_{entry}(2)$
2	$LV_{exit}(2) \setminus \{b\} \cup \{a\}$	$LV_{entry}(3)$
3	$LV_{exit}(3) \setminus \{c\} \cup \{b\}$	$LV_{entry}(4)$
4		

Example for LVA

$$
\begin{tikzcd}
{[a := 0]^1} \arrow[d] & \\
{[b := a * 1]^2} \arrow[d] & \\
{[c := c + b]^3} \arrow[d] & \\
{[a := b * 2]^4} \arrow[d] & \\
{[a < 9]^5} \arrow[uuu, bend left=60] \arrow[r] & {[\textbf{return } c]^6}
\end{tikzcd}
$$

ℓ	gen_{LV}	kill_{LV}
1	$\emptyset$	$\{a\}$
2	$\{a\}$	$\{b\}$
3	$\{b\}$	$\{c\}$
4	$\{b\}$	$\{a\}$
5	$\{a\}$	$\emptyset$

ℓ	$LV_{\text{entry}}(\ell)$	$LV_{\text{exit}}(\ell)$
1	$LV_{\text{exit}}(1) \setminus \{a\}$	$LV_{\text{entry}}(2)$
2	$LV_{\text{exit}}(2) \setminus \{b\} \cup \{a\}$	$LV_{\text{entry}}(3)$
3	$LV_{\text{exit}}(3) \setminus \{c\} \cup \{b\}$	$LV_{\text{entry}}(4)$
4	$LV_{\text{exit}}(4) \setminus \{a\} \cup \{b\}$	

Example for LVA

$[a := 0]^1$

$\downarrow$

$[b := a * 1]^2$

$\downarrow$

$[c := c + b]^3$

$\downarrow$

$[a := b * 2]^4$

$\downarrow$

$[a < 9]^5 \longrightarrow [\textbf{return } c]^6$

ℓ	gen_{LV}	$kill_{LV}$
1	$\emptyset$	$\{a\}$
2	$\{a\}$	$\{b\}$
3	$\{b\}$	$\{c\}$
4	$\{b\}$	$\{a\}$
5	$\{a\}$	$\emptyset$

ℓ	$LV_{entry}(\ell)$	$LV_{exit}(\ell)$
1	$LV_{exit}(1) \setminus \{a\}$	$LV_{entry}(2)$
2	$LV_{exit}(2) \setminus \{b\} \cup \{a\}$	$LV_{entry}(3)$
3	$LV_{exit}(3) \setminus \{c\} \cup \{b\}$	$LV_{entry}(4)$
4	$LV_{exit}(4) \setminus \{a\} \cup \{b\}$	$LV_{entry}(5)$
5		

Example for LVA

$$\begin{tikzcd}
{[a := 0]^1} \arrow[d] \\
{[b := a * 1]^2} \arrow[d] \\
{[c := c + b]^3} \arrow[d] \\
{[a := b * 2]^4} \arrow[d] \\
{[a < 9]^5} \arrow[r] \arrow[uuu, bend left] & {[\textbf{return } c]^6}
\end{tikzcd}$$

ℓ	gen_{LV}	kill_{LV}
1	$\emptyset$	$\{a\}$
2	$\{a\}$	$\{b\}$
3	$\{b\}$	$\{c\}$
4	$\{b\}$	$\{a\}$
5	$\{a\}$	$\emptyset$

ℓ	$LV_{\text{entry}}(\ell)$	$LV_{\text{exit}}(\ell)$
1	$LV_{\text{exit}}(1) \setminus \{a\}$	$LV_{\text{entry}}(2)$
2	$LV_{\text{exit}}(2) \setminus \{b\} \cup \{a\}$	$LV_{\text{entry}}(3)$
3	$LV_{\text{exit}}(3) \setminus \{c\} \cup \{b\}$	$LV_{\text{entry}}(4)$
4	$LV_{\text{exit}}(4) \setminus \{a\} \cup \{b\}$	$LV_{\text{entry}}(5)$
5	$LV_{\text{exit}}(5) \cup \{a\}$	

Example for LVA

$$
\begin{tikzcd}
{[a := 0]^1} \arrow[d] & \\
{[b := a * 1]^2} \arrow[d] & \\
{[c := c + b]^3} \arrow[d] & \\
{[a := b * 2]^4} \arrow[d] & \\
{[a < 9]^5} \arrow[r] \arrow[uuu, bend left=60] & {[\textbf{return } c]^6}
\end{tikzcd}
$$

ℓ	gen_{LV}	$kill_{LV}$
1	$\emptyset$	$\{a\}$
2	$\{a\}$	$\{b\}$
3	$\{b\}$	$\{c\}$
4	$\{b\}$	$\{a\}$
5	$\{a\}$	$\emptyset$

ℓ	$LV_{entry}(\ell)$	$LV_{exit}(\ell)$
1	$LV_{exit}(1) \setminus \{a\}$	$LV_{entry}(2)$
2	$LV_{exit}(2) \setminus \{b\} \cup \{a\}$	$LV_{entry}(3)$
3	$LV_{exit}(3) \setminus \{c\} \cup \{b\}$	$LV_{entry}(4)$
4	$LV_{exit}(4) \setminus \{a\} \cup \{b\}$	$LV_{entry}(5)$
5	$LV_{exit}(5) \cup \{a\}$	$LV_{entry}(2) \cup LV_{entry}(6)$

Results

$$[a := 0]^1$$
$$\downarrow$$
$$[b := a * 1]^2$$
$$\downarrow$$
$$[c := c + b]^3$$
$$\downarrow$$
$$[a := b * 2]^4$$
$$\downarrow$$
$$[a < 9]^5 \longrightarrow [\textbf{return } c]^6$$

ℓ	gen_{LV}	kill_{LV}
1	$\emptyset$	$\{a\}$
2	$\{a\}$	$\{b\}$
3	$\{b\}$	$\{c\}$
4	$\{b\}$	$\{a\}$
5	$\{a\}$	$\emptyset$

ℓ	$LV_{\text{entry}}(\ell)$	$LV_{\text{exit}}(\ell)$
1	$\emptyset$	$\{a\}$
2	$\{a\}$	$\{b\}$
3	$\{b\}$	$\{b, c\}$
4	$\{b, c\}$	$\{a, c\}$
5	$\{a, c\}$	$\{a, c\}$

Results

$$
\begin{tikzcd}
{[a := 0]^1} \arrow[d] & \\
{[b := a * 1]^2} \arrow[d] & \\
{[c := c + b]^3} \arrow[d] & \\
{[a := b * 2]^4} \arrow[d] & \\
{[a < 9]^5} \arrow[uuu, bend left=60] \arrow[r] & {[\textbf{return } c]^6}
\end{tikzcd}
$$

ℓ	gen_{LV}	kill_{LV}
1	$\emptyset$	$\{a\}$
2	$\{a\}$	$\{b\}$
3	$\{b\}$	$\{c\}$
4	$\{b\}$	$\{a\}$
5	$\{a\}$	$\emptyset$

ℓ	$LV_{\text{entry}}(\ell)$	$LV_{\text{exit}}(\ell)$
1	$\emptyset$	$\{a\}$
2	$\{a\}$	$\{b\}$
3	$\{b\}$	$\{b, c\}$
4	$\{b, c\}$	$\{a, c\}$
5	$\{a, c\}$	$\{a, c\}$

Note: b and a are never simultaneously live!

Existence of Solutions

Solutions **always exist** to our data flow equations.
Why? Because $(2^{\mathcal{A}}, \subseteq)$ (for AEA) and $(2^{\mathsf{Var}}, \subseteq)$ (for LVA) are both lattices and all our functions are monotone. So the Knaster-Tarski theorem applies.

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